HOMEWORK 1.3 · UNIT 1 · NC.8.EE.2 · NC.8.NS.1

Square Roots

Targets: EE.2-1, EE.2-2, EE.2-4, NS.1-1, NS.1-4, EE.2-6, EE.2-3, NS.1-6, EE.2-8

Name _____ Date _____ Block _____

Remember: A **square root** undoes squaring: $\sqrt{49} = 7$ because $7 \times 7 = 49$. A **cube root** undoes cubing: $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$. If the number under $\sqrt{\ }$ is a perfect square, the root is rational. If not, it is irrational.

SCORE 3.0 Approaching · the basics

Targets: EE.2-1, EE.2-2, EE.2-4, NS.1-1, NS.1-4

1 Find 15^2 EE.2-1

2 Find $\sqrt{144}$ and $\sqrt[3]{8}$ EE.2-2

3 Circle one: $\sqrt{30}$ EE.2-4

Rational Irrational

Why? _____

4 Which fraction makes a **repeating** decimal? REVIEW 1.1 NS.1-1

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{5}$

5 −5 is a whole number. REVIEW 1.2 NS.1-4

True False

SCORE 4.0 Proficient · on target

Targets: EE.2-6, EE.2-3, NS.1-6

EE.2-6

6 Find $\sqrt{256}$

Ⓐ 14 Ⓑ 16 Ⓒ 18 Ⓓ 128

EE.2-6,EE.2-3

7 **Nia** made a mistake. Circle it. Explain it. Fix it.

NIA'S WORK

$\sqrt{64} = 32$

because $64 \div 2 = 32$

REVIEW 1.2 NS.1-6

8 Which numbers are **rational**?

☐ $\sqrt{49}$ ☐ $\sqrt{15}$ ☐ $\sqrt[3]{27}$ ☐ π

☐ $\frac{2}{3}$

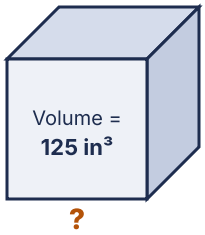
Select all that apply.

SCORE 5.0 Mastery · go deeper

Targets: EE.2-8, EE.2-3

EE.2-8

9 A cube-shaped box holds 125 in^3. How long is each edge?



EE.2-3

10 Explain why $\sqrt{9^2} = 9$. Use the words **inverse operations**.

$\sqrt{9^2} = 9$ because _____ .